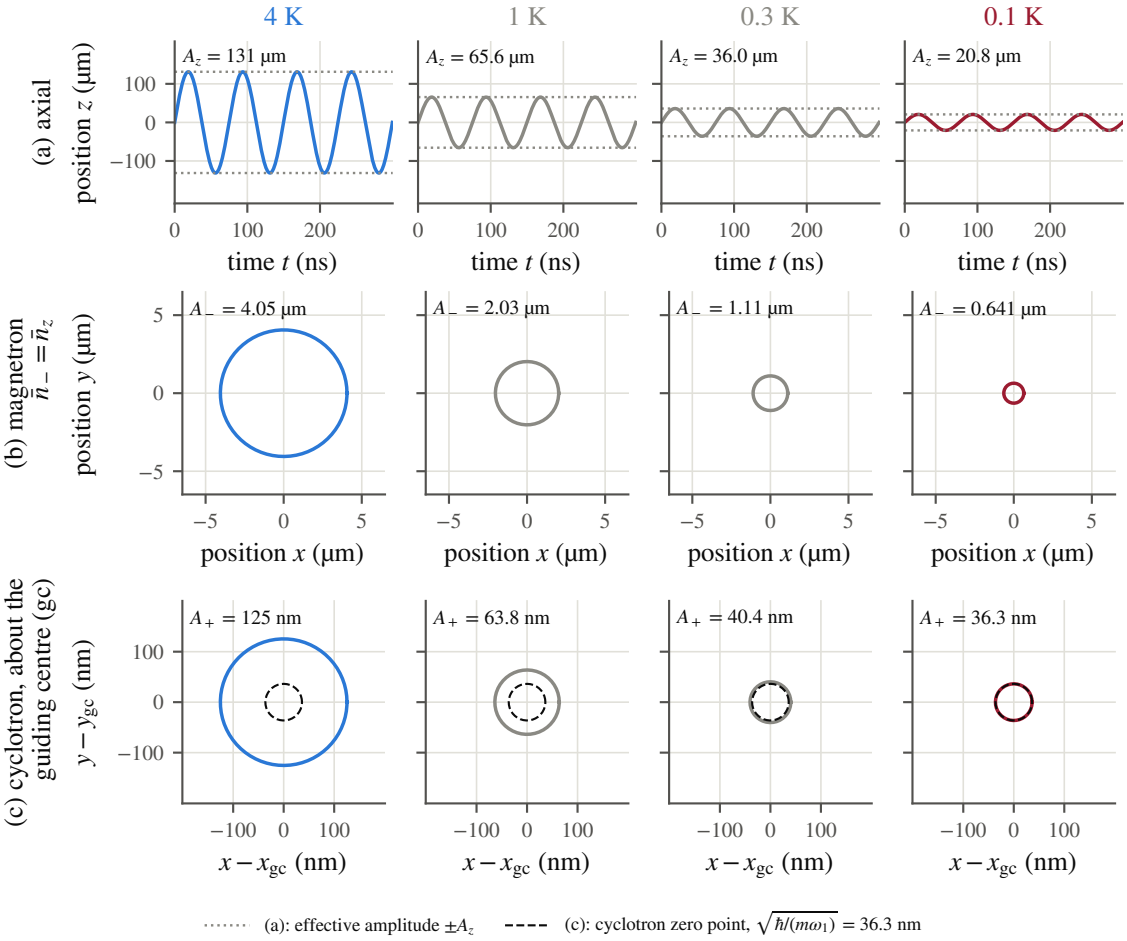
4 K
1 K
0.3 K
0.1 K
(a) axial position $z$ (μm)
$A_z$ = 131 μm
$A_z$ = 65.6 μm
$A_z$ = 36.0 μm
$A_z$ = 20.8 μm
time $t$ (ns)
(b) magnetron $\vec{n}_- = \vec{n}_z$
position $y$ (μm)
$A_-$ = 4.05 μm
$A_-$ = 2.03 μm
$A_-$ = 1.11 μm
$A_-$ = 0.641 μm
position $x$ (μm)
(c) cyclotron, about the guiding centre (gc)
$y - y_{gc}$ (nm)
$A_+$ = 125 nm
$A_+$ = 63.8 nm
$A_+$ = 40.4 nm
$A_+$ = 36.3 nm
$x - x_{gc}$ (nm)
(a): effective amplitude $\pm A_z$
(c): cyclotron zero point, $\sqrt{\hbar/(m\omega_1)}$ = 36.3 nm